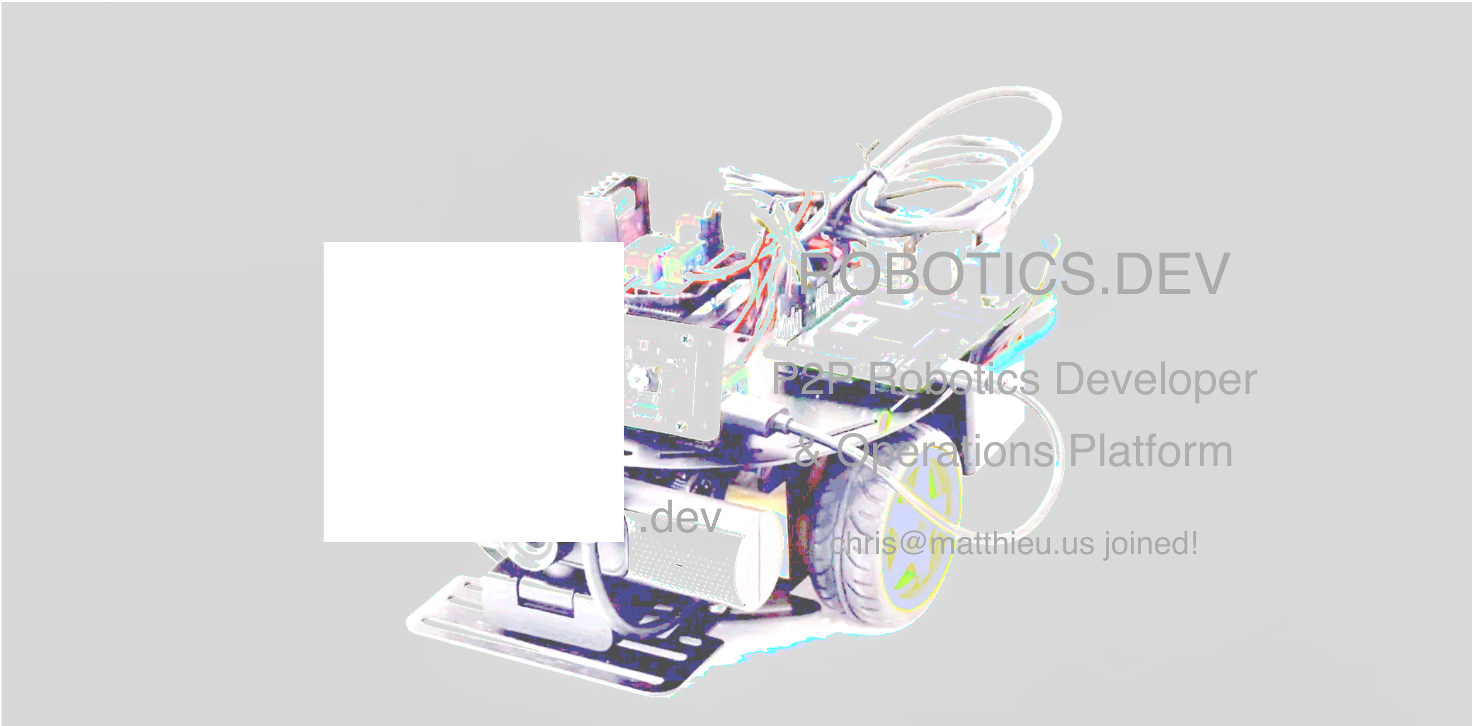
robotics.dev



Connect, Code, and Control ROS robots remotely via secure peer-to-peer (P2P) connectivity, Teleop remote control, Node.JS and Python Robotics Developer Kits (RDK), REST APIs, and VSCode/Cursor Copilot AI. Develop, test, and deploy your new Robotics.dev P2P apps anywhere (cloud, edge, or on your robots)!

Global P2P Connectivity

Extend ROS2 robotics capabilities beyond local area networks for worldwide robot control with our Web Teleop, Robotics Developer Kit, REST APIs, and Copilot AI on **Linux, Windows, and Mac!**

P2P Compute

Run AI-powered robotics apps peer-to-peer on the cloud or edge or your development environment, or even on the robot itself. Runs inside **Docker** containers anywhere too!

P2P Web Teleop

Supports Web Teleop control from anywhere in the world over browsers with P2P streaming video with both software and hardware **joysticks/gamepads**. Works great on **mobile** devices too!

Multi-Robot Orchestration

Develop and run AI tasks across fleets of robots or orchestrate multiple AI tasks on the same robot from anywhere with ease and without interrupting existing ROS apps running on the robot.

Easy Integration

Runs along side of existing ROS2 apps. Compatible with 2D and 3D depth **stereo cameras** and a wide range of hardware platforms. We also

Developer Tools

Our developer tools include:

- VSCode/Cursor Extension
- Copilot Gen-AI
- Robotics CLI

offer a powerful Robotics Developer Kit (**RDK**) as well as a simple and secure **REST API** and WebSocket support for developers!

- NodeJS P2P RDK
- Python P2P RDK
- REST APIs

Robotics.dev works out-of-the-box with any ROS2 robotics hardware.
If you need a DIY ROS2 Motor Controller, we natively support the following GPIO boards:

Raspberry Pi

RPi Zero, RPi3, RPi4, RPi5

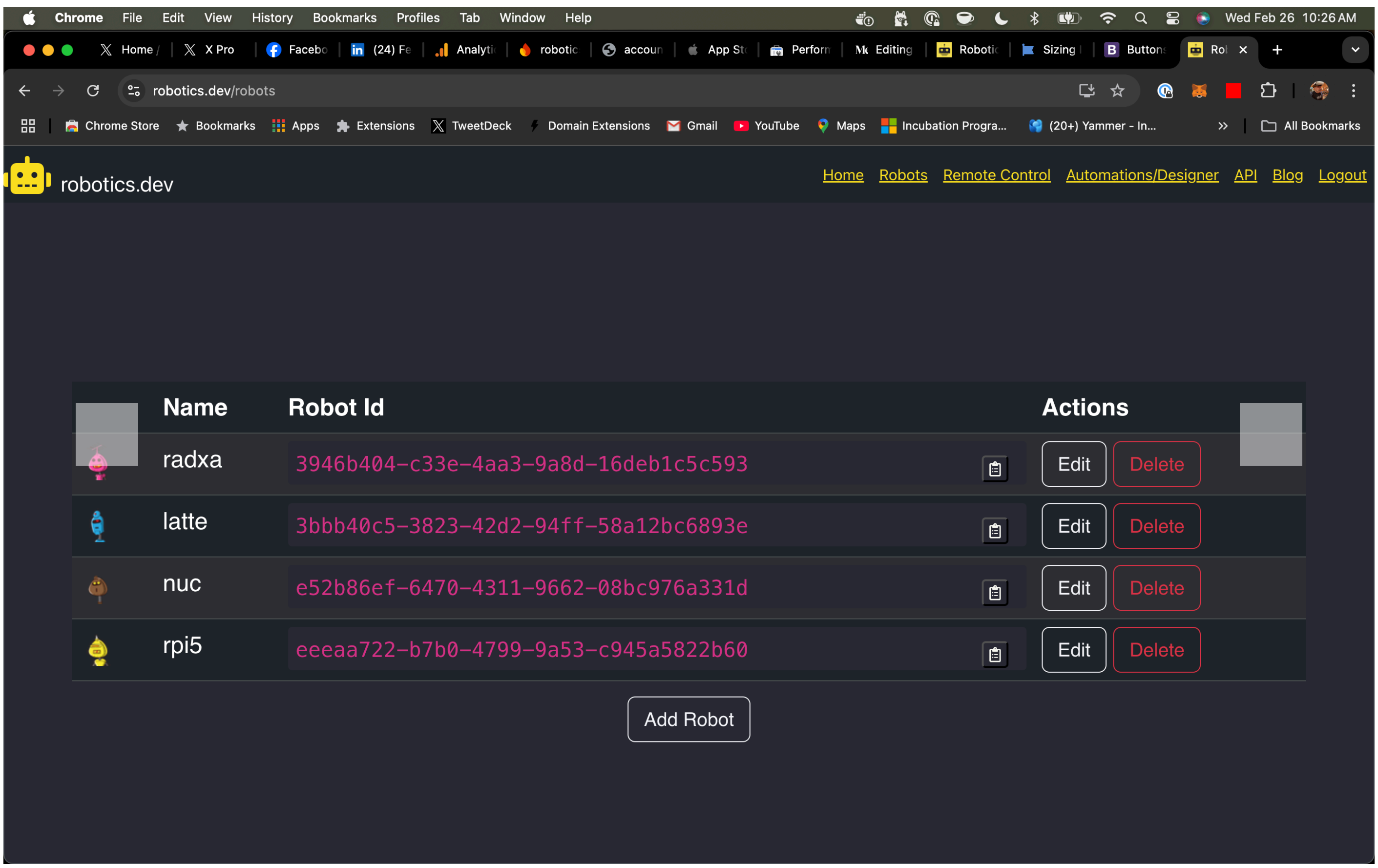
Radxa X4 (x86)

LattePanda 3 Delta (x86)

Intel NUC

with Arduino/ESP32

Sneak Peak!



How Does It Work?

Getting Started with Robot

Install Robotics.dev CLI (command line interface) on robot. Learn more about our [Robotics NPM package].

```
npm install -g robotics
```

Get Robot ID and add it to your list of robots in Robotics.dev

```
robotics id
```

Add your developer API token to your robot

```
robotics set --token=1234...6789
```

Start Communications (defaults to robotics.dev)

```
robotics connect
```

Start Motors (defaults to Raspberry Pi)

```
robotics start motors
```

Start 2D Camera (defaults to device /dev/video0)

```
robotics start camera
```

Returns help on all commands and parameters

```
robotics help
```

Getting Started with RDK (Robotics Developer Kit)

Install Robotics.dev Node.JS or Python RDK on development machine (Linux, Windows, Mac).

Node.JS Python

Learn more about our [Robotics-dev NPM NodeJS RDK].

```
npm install robotics-dev
```

```
//import robotics from 'robotics-dev';
const robotics = require('robotics-dev');

// Define ROS twist movement commands
const moveForward = {
  linear: {x: 0.2, y: 0.0, z: 0.0},
  angular: {x: 0.0, y: 0.0, z: 0.0}
};
const turnLeft = {
  linear: {x: 0.0, y: 0.0, z: 0.0},
  angular: {x: 0.0, y: 0.0, z: 0.2}
};
const turnRight = {
  linear: {x: 0.0, y: 0.0, z: 0.0},
  angular: {x: 0.0, y: 0.0, z: -0.2}
};
const stop = {
  linear: {x: 0.0, y: 0.0, z: 0.0},
  angular: {x: 0.0, y: 0.0, z: 0.0}
};

var latestImage;
var oneTime = true;
const robotId = 'ENTER ROBOTICS.DEV ROBOT ID HERE';
```

```
const apiToken = 'ENTER ROBOTICS.DEV DEVELOPER API TOKEN HERE';

// Connect RDK to robot via P2P and start listening for ROS messages
robotics.connect({robot: robotId, token: apiToken}, (ros) => {
  console.log('Received p2p data:', ros);
  // {
  //   robotId: '3bbb40c5-xxxx-yyyy-zzzz-58a12bc6893e',
  //   topic: '/camera2d',
  //   data: {
  //     data: xyz
  //   }
  // }
  if(ros.topic === '/camera2d'){
    console.log("Base64 image: ", ros.data.data);
    latestImage = ros.data.data
  }

  if(oneTime){
    oneTime = false;
    robotics.speak(robotId, 'beep boop')

    console.log('Moving robot forward at 20% speed...');
    robotics.twist(robotId, moveForward);

    // Stop after 5 seconds
    setTimeout(() => {
        console.log('Stopping robot...');
        robotics.twist(robotId, stop);
    }, 5000);
  }

});
```